

Research Article

Deep-Learning-Based Real-Time Visual Detection Method for Robotic Cell Microinjection System With High Accuracy and Efficiency

Guozhi Liu , Shengzheng Kang, Tao Li , Yifan Xu, and Jie Zhou

School of Automation, Nanjing University of Information Science and Technology, Nanjing 210044, Jiangsu, China

Correspondence should be addressed to Tao Li; litaojia@nuist.edu.cn

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The advancement of the life sciences has led to an increased demand for biological cell microinjection techniques. A number of algorithms have been developed for the purpose of detecting cells, and employed in automated cell microinjection systems. However, due to the significant difference in scale between the microinjector tip size ($\sim 10\ \mu\text{m}$) and the cell size ($\sim 100\ \mu\text{m}$), conventional cell detection algorithms struggle to accurately detect both the microinjector tip and the cell simultaneously. Additionally, both the cells and the microinjector are transparent, making it difficult to distinguish them from the culture medium under the light. This leads to a lack of real-time coordinate feedback for the cell and microinjector tip, increasing the risk of damaging the cell during puncture. To this end, this paper proposes a deep-learning based real-time visual detection method for robotic cell microinjection system with high accuracy and efficiency. The core of the method is the target detection algorithm based on improved YOLOv8n (denoted as BI-YOLO). The deformable convolution is added to the backbone of algorithm to adapt to different shapes and sizes of cells and microinjector tips. And bidirectional feature pyramid network (BiFPN) which adds adaptive feature fusion is used as the neck part of the algorithm to achieve higher accuracy and precision for cell and microinjector tip detection using smaller computational resources. Zebrafish embryo injection experiments are conducted using the designed robotic micromanipulation system to verify the effectiveness of the proposed visual detection method. The experimental results show that the BI-YOLO algorithm achieves an average intersection over union (IoU) accuracy of 99.5% for cells and 97.8% for microinjector tips, with just 1.96 million parameters and 7.2 giga floating point operations per second (GFLOPs) of computation. Its overall performance surpasses other popular deep learning algorithms. The success rates for cell and microinjector tip detection are 100% and 94.7%, respectively, while the detection speed is 118.8 frame rate per second (FPS). The visual detection method enables real-time detection of positions of cell and microinjector tip and provide visual feedback. The method will be promising in the application of robotic batch cell microinjection with high efficiency.

Keywords: cell microinjection; deep learning; visual detection; YOLOv8n

1. Introduction

Microinjection is a form of cell micromanipulation, a technique employed in biological research to introduce foreign substances into biological samples. It is a common procedure utilized in various fields, including vaccine development, drug screening, and genetic research [1]. The traditional method for cell microinjection tasks involves manual execution by professionals, leading to inconsistencies in both efficiency and repeatability.

With the increasing demand in life sciences and advancements in industrial automation, the requirements for cell microinjection are progressively evolving towards greater automation and higher quality. As a result, significant efforts have been made in the system design [2], control [3–5], and visual detection [6, 7] of microinjection. Additionally, a issue that gradually emerges is how to reduce the cell damage to ensure cell survival during injection.

To minimize the damage to cells that can occur during automated injections, a number of devices and algorithms have been designed to measure and control the puncture force and grasping force during injections.

Shan et al. [8] present a serial two-degree-of-freedom compliant constant force microgripper to realize large constant force range of microgrippers. Cell mechanical properties are also gradually being taken into consideration. In [9], a new force-sensitive microinjector based on a fiber Bragg grating (FBG) sensor was designed to sense the three-dimensional force of microinjection. A new fractional model was proposed to accurately describe the viscoelastic mechanical properties of cells to simultaneously characterize the power-law relaxation and creep behavior of cells in microinjection with higher accuracy and fewer parameters in [10]. A speed optimization method is also developed based on this model to achieve minimal cell damage. To simplify the understanding of cell state, a micromanipulation interface has been proposed that measures the degree of cell deformation in real time and provides surface tactile sensation through a fabric actuator [11]. In [12], a new rate-dependent mechanical model based on membrane theory is proposed for the first time. This model establishes an analytical equilibrium equation between the injection force and cell deformation by taking into account the speed effect of microinjection. In summary, both high-quality injections and mechanical characterization of cells require a high-precision microscopic vision device and efficient detection algorithm for visual detection of cells and microinjectors.

Traditional image processing algorithms, such as thresholding, region-based methods, template matching, and Hough transform, have been used for cell detection [1, 13]. In [6], image binarization is achieved by using grayscale thresholding to distinguish the zebrafish from its image background. Then filtering, binarization, image segmentation, and angle correction are employed to obtain a clear image of standard zebrafish larvae, and finally the edge features are utilized to determine the location of injected target. A novel visual detection algorithm based on automatic thresholding and excessive dilatation was introduced to accurately identify the centers of zebrafish embryos and larval yolks in [7]. Despite the fact that these methods are capable of roughly identifying cells, they are susceptible to air bubbles, contaminants, and other factors in the dynamic system of cell microinjection, which in turn leads to experimental failures [14]. Furthermore, conventional image processing algorithms remain inadequate in terms of detection efficacy and accuracy. Consequently, the advancement of target detection through deep learning offers novel avenues for research in this domain. To achieve high-throughput injection while maintaining high injection accuracy, a visual processing technique based on convolutional neural networks was proposed for biological sample manipulation and microinjection devices [15]. Pan et al. [16] employed a deep convolutional neural network, namely Mask R-CNN, for the purpose of detecting and segmenting stain-free adherent cells. This approach has proven highly effective in improving the computational efficiency and increasing the throughput of the injection system. A combination of deep learning-based faster R-CNN target detection and traditional image processing

algorithms was employed to accurately determine the positions of the injected target and the center of the yolk in [2]. Although deep learning-based visual detection has improved the accuracy and efficiency of cell microinjection, most R-CNN-based visual detection methods operate at low speeds, making it challenging to meet the demands of real-time target detection. With the introduction of the self-attention mechanism, transform has gradually been applied in image detection [17] and visual control [18].

The advancement of deep learning technology and the growing intricacy of real-world scenarios have led to the emergence of an array of models based on diverse deep learning algorithms in the domain of target detection. These models have demonstrated a remarkable capacity to address complex problems [19–22]. Traditional deep learning-based target detection techniques are mainly based on the two-stage approach of convolutional neural networks. Subsequently, scientists developed one-stage algorithms. In general, one-stage detection algorithms are faster, while two-stage detection algorithms are more accurate. Given the necessity for real-time monitoring of cell and microinjector positions in cell microinjection, it is evident that faster one-stage detection algorithms are better suited to the harsh requirement in practice. Redmon et al. [23] put forth a one-stage algorithm, exemplified by YOLO, as a means of achieving real-time target detection results for an end-to-end deep learning system. The primary concept of the YOLO series of algorithms is to directly obtain the class and precise position of the target object from the input image, without generating candidate regions as done in the R-CNN series. This approach directly results in faster processing speeds. In [24], an improved YOLO-V4 deep learning algorithm is proposed for detecting cherry fruits. In [25], a new real-time face detector, YOLO-FaceV2, was introduced based on the YOLOv5 architecture to address issues related to face occlusion. The YOLO series has recently introduced YOLOv8, which has been widely adopted in various fields due to its high efficiency and robust performance, spanning autonomous vehicle detection [26], small target detection in aerial imagery captured by unmanned aerial vehicles (UAVs) [27], road defect detection [28], and agricultural crop detection [29]. The YOLO algorithm is also increasingly used for visual detection of cell microinjection. Sadak et al. [30] employed YOLOv2_Resnet-50 to detect zebrafish cells, achieving high real-time performance and accuracy in cell detection. However, the detection of microinjector tip employs algorithms, such as constrained adaptive histogram equalization, Gaussian filter, adaptive thresholding method, Canny edge detection, which exhibit low efficiency and real-time performance. To this end, an automatic microinjection assistance system based on deep-learning visual detection with YOLOv8n was designed to detect cells and microinjector tips in real time [31]. However, the detection algorithm of the system is unable to adaptively adjust the receptive field due to the significant scale difference between the microinjector tip size ($\sim 10\ \mu\text{m}$) and cell size ($\sim 100\ \mu\text{m}$), leading to a notable decrease in the accuracy of microinjector tip detection.

Therefore, this paper proposes a visual detection method based on an improved YOLOv8n (denoted as BI-YOLO) for an

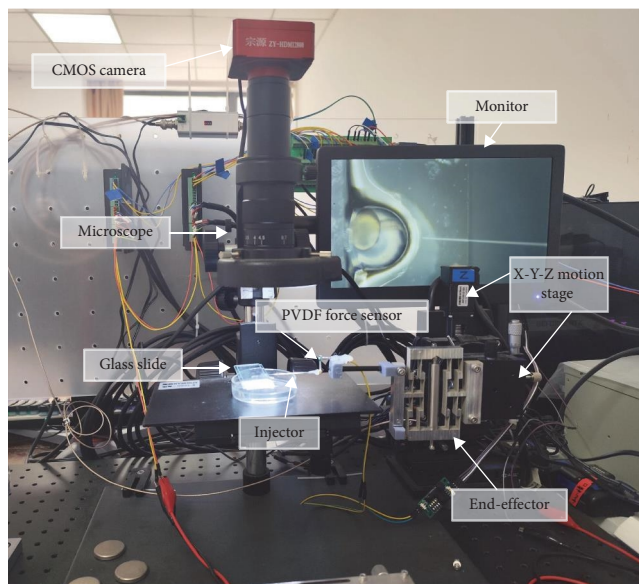


FIGURE 1: The hardware setup of the system.

automated cell microinjection system. The objective is to improve the detection accuracy of cell and microinjector tip while maintaining a high level of real-time performance. The BI-YOLO, compared to the YOLOv8n, adds deformable convolution to the backbone, which enables the algorithm to adaptively change the sampling position and accurately capture the main features when facing objects of different sizes. Its neck structure is also replaced with a lighter BiFPN structure with adaptive feature fusion, which improves the accuracy while decreasing the number of parameters and computation of the model. This visual detection method enables rapid and accurate detection of targets in microscopic images, and allows for quick acquisition of coordinate position information for cells and microinjector tips. The method helps to improve the automation of detection. Finally, experiments are performed on zebrafish embryos to validate the effectiveness of the visual detection method in cell microinjection.

The rest of the paper is organized as follows: Section 2 shows the system setup, an overview of the visual detection method and the relevant preparations for model training. Section 3 describes the principles of the improved cell and microinjector tip detection algorithms. In Section 4, experiments are conducted on zebrafish embryos to validate the effectiveness of the visual detection method in microinjection, followed by a discussion. Finally, Section 5 summarizes the paper and provides an outlook for future work.

2. System Setup and Related Preparations

2.1. System Setup. The hardware setup of the system is shown in Figure 1. The hardware implementation is mainly divided into microscopic image acquisition part, movement implementation part, main control processing part, and cell immobilization platform. The microscopic image acquisition part includes a optical microscope (model: ZY-HDMI2800), a CMOS camera, and a monitor to capture the cell microinjection motions. The movement implementation part employs a combination of

a macroactuated platform and an end-effector. The end-effector is comprised of a piezoelectric actuator and a compliant displacement amplification mechanism. The proximal end of the microinjector is rigidly mounted onto a force sensor, which is integrated into the distal terminus of a compliant mechanism-based positioning stage. The macro-actuated platform provides large-range displacement motion, whereas the end-effector assembly achieves micrometer-scale positioning accuracy for cellular penetration. This dual-stage architecture optimizes operational efficiency by decoupling coarse alignment from precision maneuvers, critical for maintaining cell viability during membrane penetration procedures. The main control processing part is constituted by a computer and a data acquisition card (NI: PCI-6229). The cell immobilization platform includes a petri dish and slides for placement and fixation of cells.

2.2. Method Flow. The flowchart for the visual detection method is shown in Figure 2. To capture a clear image for dataset recording to train the target detection model, the focus of the microscope is first adjusted to ensure that both the cell and the microinjector tip are visible. A model for cell and microinjector tip detection and localization is obtained by using the BI-YOLO algorithm designed in this paper and the generated dataset for training. The model is utilized to obtain information about positional coordinates of the cell and the microinjector tip from the captured images. These coordinates are then sent to the host computer, which is responsible for planning the trajectory of the microinjector tip. Combined with the position information, the movement implementation part is responsible for driving the microinjector to finish the injection.

2.3. Data Collection and Labeling. As a widely used class of cell models in biomedical research, zebrafish embryos are introduced in this study because of their advantages of transparency and larger size [14]. The zebrafish embryo and microinjector tip dataset is produced by Lableimg, a graphical image labeling tool. Then a dataset module for detecting zebrafish embryos and microinjector tips is trained using a computer. The internally constructed zebrafish embryo dataset comprises 1134 annotated images, containing 1454 cell samples and 958 microinjector tip samples. The dataset was partitioned into 70% for training, 20% for validation, and 10% for testing, with stratification to maintain sample type distribution across splits. Considering the size of our dataset, this ratio is sufficiently practical to demonstrate the performance and generalization ability of the trained neural network. The validation set is used to evaluate the trained neural network model. The test set is employed to finally evaluate the generalization ability of the model.

2.4. Description of YOLOv8 Algorithm. YOLOv8 is a target detection algorithm proposed by ultralytics based on the improved version of the previously released YOLOv5. The input side of YOLOv8 uses a mosaic dataset enhancement method, where four images are randomly scaled, randomly cropped, and randomly aligned, thereby enriching the dataset and reducing GPU requirements. The backbone of YOLOv8 retains the concept of CSP, but replaces the C3 module from

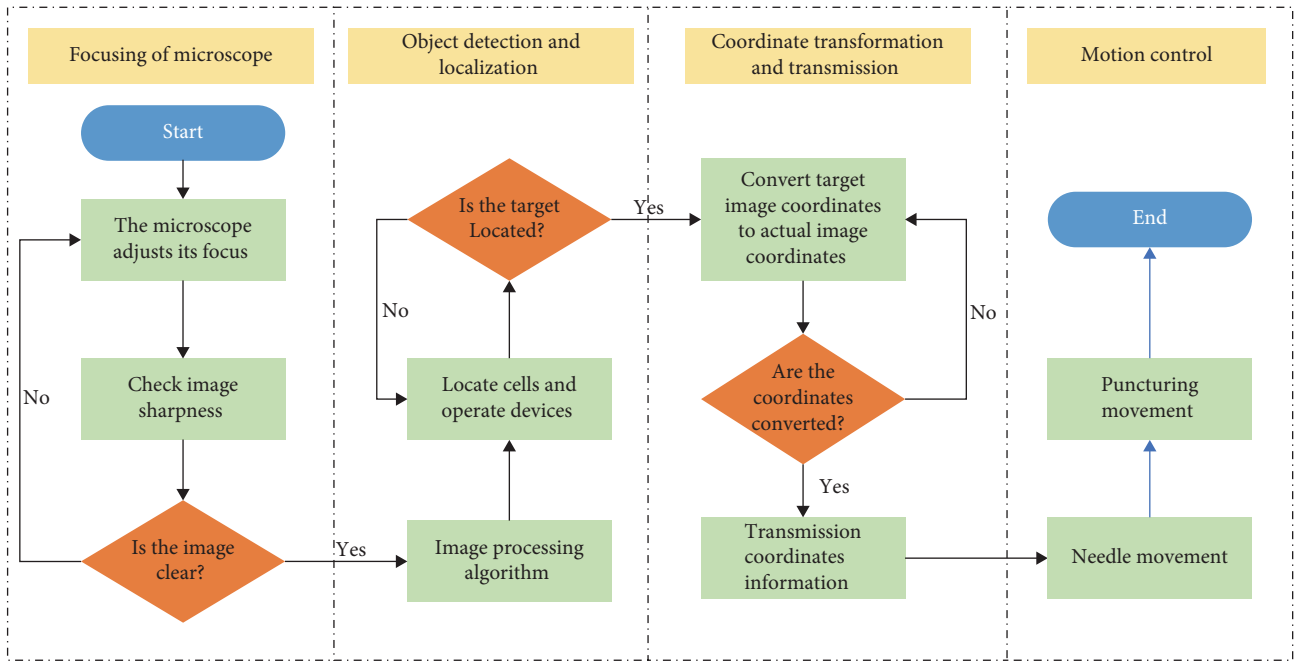


FIGURE 2: Flowchart of the visual detection method.

YOLOv5 with the C2f module, which features a richer gradient flow. This change further reduces the weight of the model. YOLOv8 continues to utilize the SPPF module that was used in YOLOv5 and other architectures. The neck of YOLOv8 continues to employ the concepts of PAN [32] and FPN [33], while removing the convolutional structure present in the upsampling stage of PAN and FPN in YOLOv5. Additionally, the head of YOLOv8 is replaced with a contemporary mainstream decoupled head structure that separates the classification and detection components, employing an anchor-free approach rather than an anchor-based one. The previous intersection over union (IoU) matching or unilateral proportion allocation has been discarded, and task-aligned assigner is now employed to match positive and negative samples. VFL loss is utilized as the classification loss, while sum of DFL loss [34] and CIOU loss [35] is used as the classification loss. The YOLOv8 algorithm identifies objects within an image and outputs the coordinates of bounding box, the category probabilities, and the confidence scores for each anchor point. The coordinates of bounding box comprise the x , y , w , and h parameters. The x and y values indicate the horizontal and vertical coordinates of the central pixel within the bounding box relative to the grid cell boundary. The w and h values represent the width and height of the bounding box in relation to the entire image. The category probability serves to identify the category to which the target in the current grid cell may be ascribed. The confidence score indicates the degree of certainty that the object in question exists within the specified grid cell. The YOLOv8 has five main versions, that is, YOLOv8n, YOLOv8s, YOLOv8m, YOLOv8l, and YOLOv8xl. For these versions, their parameter and calculation cost are increased progressively. In consideration of the computational resources and the speed of detection, this paper selects YOLOv8n, which

has the fewest parameters and computational cost, as the baseline for the BI-YOLO proposed in this paper.

2.5. Metrics. Precision, recall, mean average precision (mAP), mAP at the $\alpha\%$ IoU threshold (mAP α), mean value of the mAP calculated at 10 IoU thresholds within the interval (0.5 and 0.95) at intervals of 0.05 (mAP50-95) and detection speed (average frame rate per second [FPS]) are employed to evaluate the detectors. In addition, model size (parameters) and giga floating point operations per second (GFLOPs) are introduced to evaluate the model complexity.

$$P = \frac{TP}{TP + FP}, \quad (1)$$

$$R = \frac{TP}{TP + FN}, \quad (2)$$

$$AP = \int_0^1 P(R) dR, \quad (3)$$

$$mAP = \frac{\sum_{i=1}^k AP_i}{k}, \quad (4)$$

where P represents the precision, R represents the recall, TP represents the number of true positive samples, FP represents the the number of false positive samples, FN represents the number of false negative samples, and $P(R)$ represents the curve drawn by precision and recall.

2.6. Implementation of Data Training. The experiments described in this paper are carried out on a computer with

the Windows 11 operating system, an Intel i7-13700H CPU, and a NVIDIA 3050 GPU. The specific software environment is Python 3.10, CUDA version 11.7, and PyTorch version 2.0.0. The hyper-parameters of the BI-YOLO model training are set as follows: the initial learning rate is set to 0.01, the weight decay coefficient is 0.0005, the momentum parameter is 0.937, the number of iterations is 300, and the batch size is 16. In the initial warm-up phase, the learning rate is set to 0.1, and the warm-up momentum is set to 0.8. The training protocol incorporated mosaic data augmentation (with a probability of 0.8) for all epochs except the last 10 iterations, where it was disabled to allow model convergence without synthetic artifacts. The BI-YOLO network is constructed using the PyTorch framework, and a convolutional network is employed to facilitate the deep learning detection of the zebrafish embryo and the microinjector tip. The final training and testing are conducted on the PyCharm platform.

3. Design of BI-YOLO Object Detection Algorithm

The success rate of microinjection in robotic injection systems largely depends on the efficiency and accuracy of target detection. In contrast to standard industrial and mechanical components, zebrafish embryos exhibit distinct morphological differences. Additionally, impurities and air bubbles in the petri dish can impede image processing methods, representing a significant challenge in enhancing the success rate of microinjection. To address these issues, this paper employs deep learning-based YOLOv8n as the baseline for a visual detection algorithm and improve upon it, proposing a novel algorithm called BI-YOLO. The backbone of BI-YOLO retains the concept of CSP, but introduces deformable convolution into the C2f module, thereby enhancing the capacity to extract features of varying scale in zebrafish embryos and microinjector tips. Additionally, this integration reduces the number of model parameters. The neck of BI-YOLO abandons the original PAN + FPN structure of YOLOv8n and adopts the bidirectional feature pyramid network (BiFPN) structure [36] with adaptive feature fusion. This is a streamlined and efficient weighted BiFPN, which significantly reduces the number of parameters and the computational cost while maintaining accuracy. Figure 3 illustrates the overall detailed structure of BI-YOLO.

The C2f module splits the input feature map into two parallel branches. The first branch processes features through multiple residual units (bottleneck) to enhance network depth and feature extraction capability, while the second branch applies convolutional processing. Subsequently, the outputs from both branches are concatenated and fused via a convolutional layer for channel-wise feature integration. The C2f_DCN module replaces all standard convolutions in the C2f architecture with deformable convolution. The CBS module constitutes a fundamental unit across the network and is composed of a convolutional layer, a batch normalization layer, and a SiLU activation function layer. The SPPF module divides the input feature map into two paths. The primary path sequentially passes through three parallel max-pooling layers, while the

secondary path retains original features via direct convolutional processing. The concatenated outputs are then fused through a 1×1 convolutional layer, effectively reducing computational complexity and parameter count compared to conventional SPP modules. For detection heads, features are extracted using two CBS modules with 3×3 convolutions and one additional 1×1 convolution for final information aggregation. These processed features are then used to compute the bounding box loss and classification loss.

3.1. Deformable Convolution. Due to the changing morphology of zebrafish embryos over time, embryos with different shapes may appear in various locations in the field of view (FOV). Additionally, since cells and microinjector tip vary greatly in size in the same FOV, adaptive determination of the scale or receptive field size is required for visual detection with precise positioning. To address this, the deformable convolution [37] is introduced into the BI-YOLO in this paper. The offset is generated by applying a convolutional layer to the same input feature map, and it is then added to the original convolutional sampling position. This allows the algorithm to adaptively change the sampling position when facing objects of different sizes, accurately capturing main features and improving the precision of cell and microinjector tip detection. In Figure 4, the deformable convolution concept is illustrated.

General 2D convolution involves two steps: first, sampling from the input feature map x using a regular grid $\mathcal{M}$; second, summing the sampled values weighted by w . The grid $\mathcal{M}$ defines the receptive field size and dilation. For example,

$$\mathcal{M} = \{(-2, -2), (-2, 0), \dots, (0, 2), (2, 2)\}, \quad (5)$$

defines a 3×3 kernel with dilation 2.

For every position q_0 on the output feature map y :

$$y(q_0) = \sum_{q_n \in \mathcal{M}} w(q_n) \cdot x(q_0 + q_n), \quad (6)$$

where q_n enumerates the positions in $\mathcal{M}$.

In deformable convolution, the regular grid $\mathcal{M}$ is represented by an offset $\{\Delta q_n | n = 1, \dots, N\}$ for enhancement, where $N = |\mathcal{M}|$. Equation (6) becomes:

$$y(q_0) = \sum_{q_n \in \mathcal{M}} w(q_n) \cdot x(q_0 + q_n + \Delta q_n). \quad (7)$$

Now, the sampling is at the irregular and offset positions $q_n + \Delta q_n$, where Δq_n is the offset obtained by the convolutional layer applied on the feature map, as shown in the offsets in Figure 3.

In this paper, the C2f of the ninth layer of YOLOv8n backbone network (as P9 in the backbone of Figure 3) is improved by adding deformable convolution into it. The inclusion of deformable convolution in the low-resolution feature layer is more effective than in the high-resolution feature layer, because the deeper features mainly represent the high-level features of target. The microinjector tip is a small target that has less semantic information and requires the help of contextual

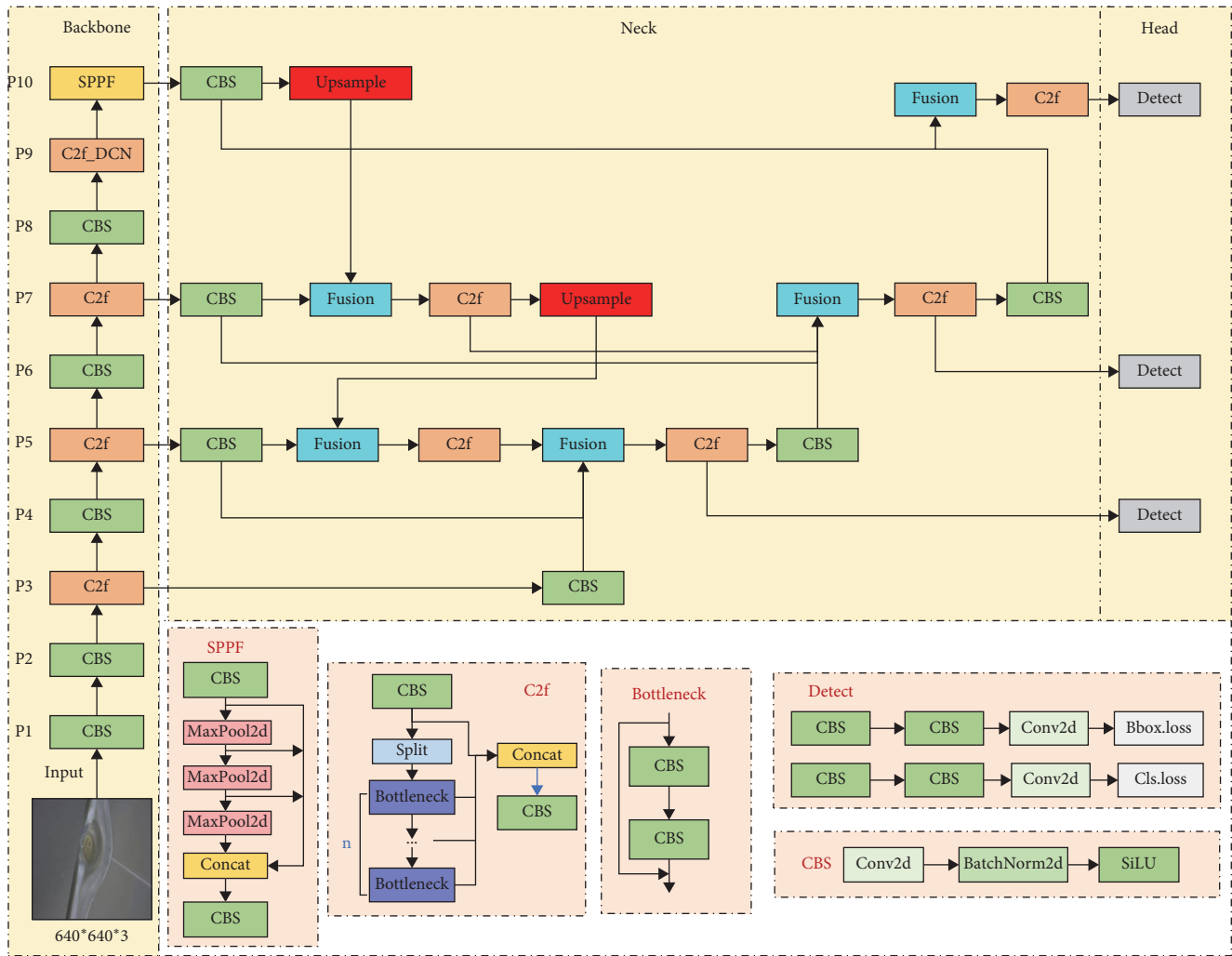


FIGURE 3: Overall structure of BI-YOLO.

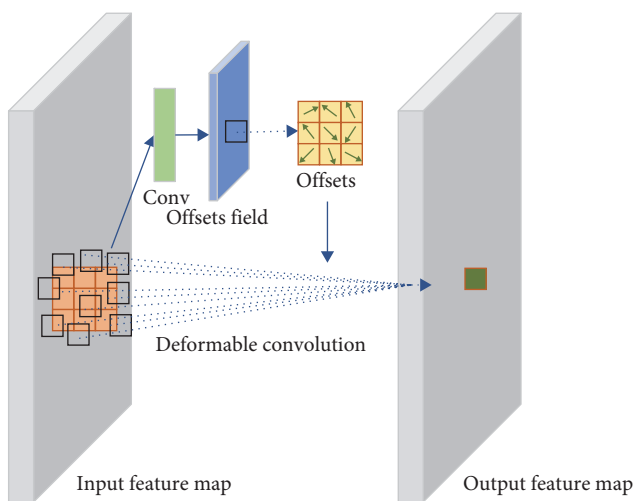


FIGURE 4: Schematic diagram of deformable convolution principle.

information for detection. On the other hand, high-level features are a collection of its own features and surrounding features, and have advanced semantic information. Therefore,

high-level features are crucial for accurately detecting small targets, and it is more efficient to add sampling offsets to this feature layer. The results in Table 1 also confirm the aforementioned statements.

3.2. Adaptive Feature Fusion. A common way to fuse features of different resolutions in a feature fusion network is to adjust them to the same resolution and then sum them. Common feature fusions are weighted fusion, cascade fusion, etc. Cascade fusion [38] is a sequential or hierarchical fusion method for situations that require sequential dependencies or hierarchies between models, and the structure is not flexible enough to adapt to rapidly changing data or task requirements. Weighted fusion [36] is a linear fusion method that works by assigning different weights to the outputs of different models and then summing the weighted outputs. It is suitable for the case where the model outputs can be summed directly. However, when various input features are at varying resolutions, their contributions to the output features are usually unequal, and the fixed weights cannot capture the complex nonlinear relationship between the model outputs. To this end, this paper introduces an adaptive feature fusion [39] in BiFPN to address the above

TABLE 1: Performance of the model with deformable convolution added at different locations in the backbone.

Location	mAP50		mAP50-95	
	Cell (%)	Microinjector tip (%)	Cell (%)	Microinjector tip (%)
DC_3	99.5	95.1	95.7	49.4
DC_5	99.5	96.5	94.8	50.1
DC_7	99.5	96.3	94.9	50.5
DC_9	99.5	97.7	95.0	51.9

Note: 3, 5, 7, and 9 represent the layers in the backbone network. Bold values indicate the highest scores in each column, showing that DC_9 performed best overall. Abbreviation: DC, deformable convolution.

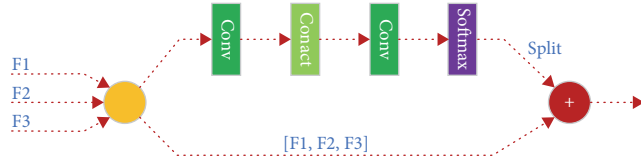


FIGURE 5: Adaptive feature fusion.

problem. Assuming the input size is represented as (bs, C, H, W) , the spatial adaptive weights of $(bs, 3, H, W)$ are derived through convolution, concatenation, and softmax operations. Each of the three channels corresponds sequentially to the three inputs. By computing the weighted sum, contextual information can be consolidated into the outputs. The structure is shown in Figure 5.

3.3. BiFPN. High-level characteristics offer more precise information and assist in the detection of small targets. For large instance detection, low-level features offer a larger FOV and more robust semantic information. By fusing high-resolution and low-resolution feature layers, the feature pyramid network is able to accurately predict targets of different sizes at different scales, thus enhancing the performance of the model [33]. However, the lengthy path from the low-level features to the top-level features makes it more difficult to obtain precise localization data. Thus, in order to enhance the performance of the algorithm, BiFPN is introduced in this paper. BiFPN performs path augmentation and aggregation to shorten the information path and improve the feature pyramid by using the accurate positioning signals present in the lower layers, and creates bottom-up path augmentation on top of the FPN, which makes it easier to propagate the information in the lower layers. BiFPN also suggests a number of cross-scale connection optimizations to boost the effectiveness of model. First, nodes with a single input edge are eliminated. Because if a node has only one input edge without feature fusion, it contributes less to a feature network designed to fuse different features. This leads to a simplified bidirectional network. Second, BiFPN adds an additional edge at the same level from the original input to the output node, allowing for the fusion of more features without incurring excessive costs [36]. The structural sketch is shown in Figure 6b.

After the image is extracted by the BI-YOLO backbone, eight main feature layers (P3 ~ P10 in Figure 3) with progressively higher semantic information are obtained, and the feature layers that have been processed by three C2f and SPPF

(P3, P5, P7, and P10) are selected to be handed over to the BiFPN for further processing. The BiFPN integrates bidirectional cross-scalar connectivity and adaptive fusion for continuous feature fusion to further enhance the feature extraction. Using the layer 7 (P7) feature fusion as an example, Figure 6a displays the original FPN and PAN structure, which was synthesized as follows:

$$P_{10}^{td} = \text{Conv}(P_{10}^{in}), \quad (8)$$

$$P_7^{td} = \text{Conv}(P_7^{in} + \text{Resize}(P_{10}^{td})), \quad (9)$$

$$P_7^{out} = \text{Conv}(P_7^{td} + \text{Resize}(P_5^{out})). \quad (10)$$

The modified BiFPN structure was synthesized in the following form:

$$P_7^{td} = \text{Conv}\left(\frac{\theta_1 \times P_7^{in} + \theta_2 \times \text{Resize}(P_{10}^{in})}{\theta_1 + \theta_2 + \sigma}\right), \quad (11)$$

$$P_7^{out} = \text{Conv}\left(\frac{\theta'_1 \times P_7^{in} + \theta'_2 \times P_7^{td} + \theta'_3 \times \text{Resize}(P_5^{out})}{\theta'_1 + \theta'_2 + \theta'_3 + \sigma}\right). \quad (12)$$

where P_7^{in} is the input feature of layer 7. P_7^{td} is the intermediate feature at layer 7 in the top-down path. P_7^{out} is the output feature of layer 7 in the bottom-up path. θ_i and θ'_i denotes the learnable weights in the top-down and bottom-up paths, respectively, obtained by adaptive feature fusion in the previous subsection, and σ is a very small value to prevent numerical instability. “Conv” is a standard convolutional operation and “Resize” is an upsampling or downsampling operation for resolution matching [40].

4. Experimental Results and Discussion

4.1. Detection Experiment on Zebrafish Embryo and Microinjector Tip. Figure 7a is an image in the FOV captured by the CMOS camera and optical microscope, which is fed into the BI-YOLO algorithm to detect the target, and the output result is an image of the prediction frames of the zebrafish embryos and the microinjector tip with the center coordinate value, as shown in Figure 7b. In the detection result image, different colored detection boxes are used to correspond to the positions of the cell and the microinjector tip. The number

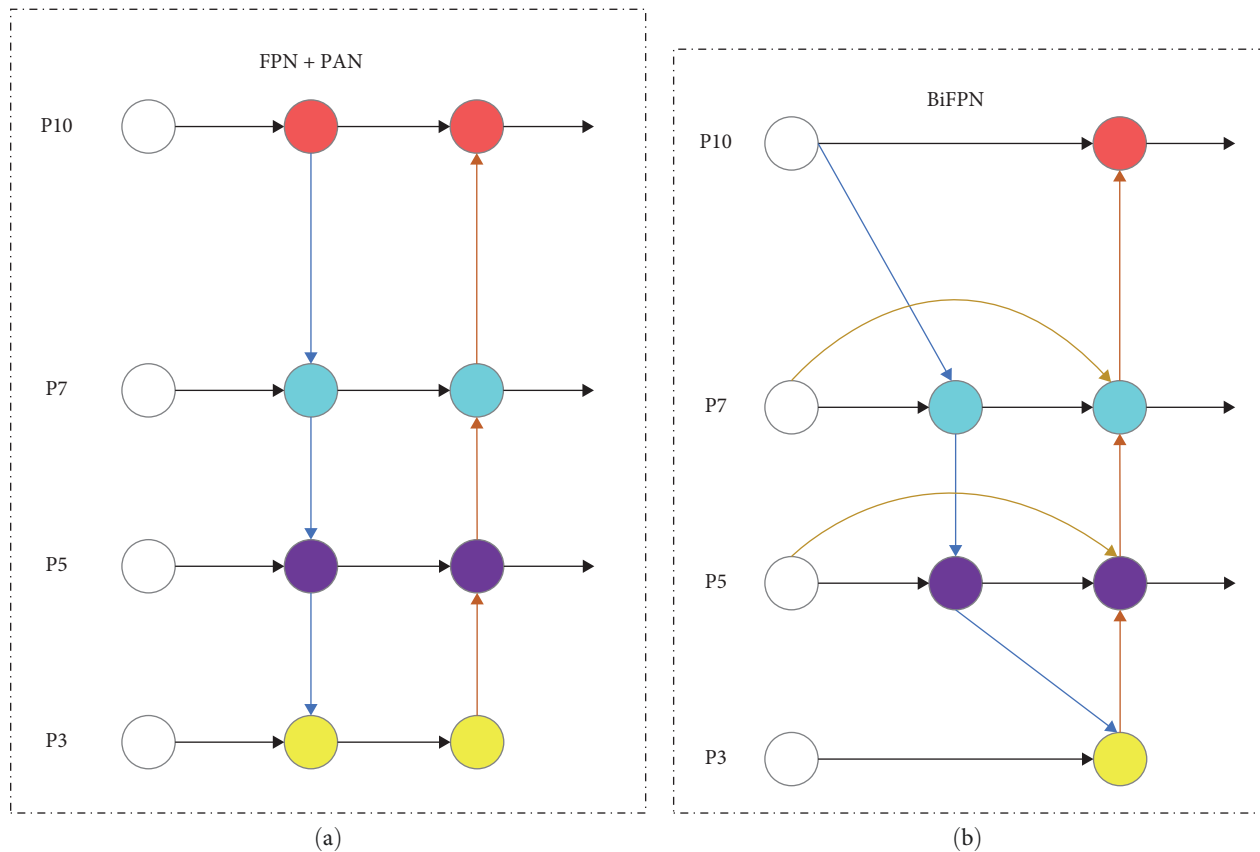


FIGURE 6: Two different neck structures: (a) FPN+PAN and (b) BiFPN.

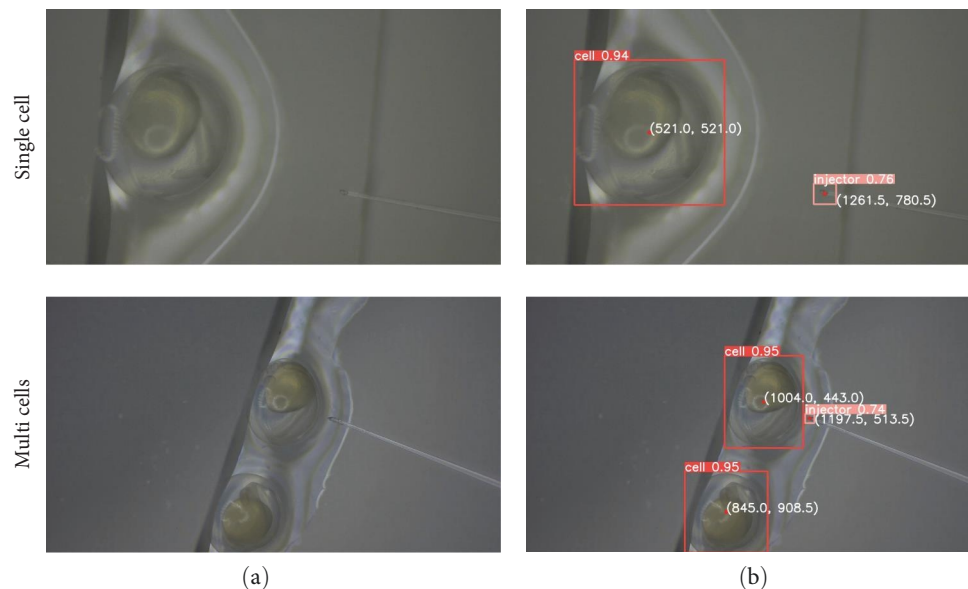


FIGURE 7: (a) Original image. (b) Image of the detection result with the center coordinate value.

on the prediction box is the confidence level of the detection result. As illustrated in Figure 7, irrespective of the position of the target within the image, the BI-YOLO algorithm not only accurately identifies single or multiple cells and microinjector

tips, but also precisely locates the coordinates of each cell and microinjector tip.

As cells are relatively fragile, excessive movement of the microinjector tip after puncturing the cell can easily damage

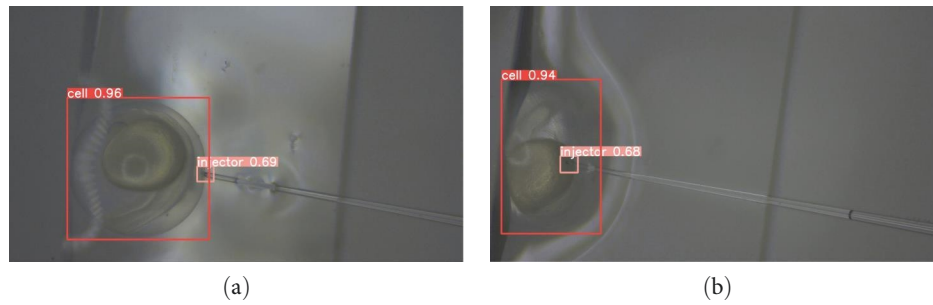


FIGURE 8: Detection of cells and microinjector tip before and after puncturing cells: (a) before puncturing the cell and (b) after puncturing the cell.

TABLE 2: The mAP of BI-YOLO for different IoU thresholds and detection speed.

Model	mAP50		mAP60		mAP70		FPS
	Cell	Microinjector tip	Cell	Microinjector tip	Cell	Microinjector tip	
BI-YOLO	99.5%	97.8%	95.4%	90.2%	99.5%	69.5%	118.8

TABLE 3: Comparison with existing studies on zebrafish detection.

Reference	Model	Success rate		FPS
		Cell (%)	Microinjector tip	
[30]	YOLOV2-Resnet-50	89.0	—	33.0
[41]	YOLOv7	95.54	—	—
[42]	Inception v3	93.0	—	—
Ours	BI-YOLO	100	94.1%	118.8

Note: Bold values indicate the highest scores in each column, showing that the algorithm of this article performed best overall.

the cell. Therefore it is essential to monitor the position of the microinjector tip both before and after it enters the cell, in order to facilitate the control of the termination or continued movement of the microinjector. Nevertheless, following the introduction of the microinjector into the cell, it is susceptible to interference or obscuration by the internal cellular structure, which is frequently not identified by general target detection algorithms. In this paper, the BI-YOLO algorithm is capable of identifying microinjector in scenarios where they are partially occluded. Furthermore, it can promptly identify microinjector tip that have entered a cell and immediately issue a stop or withdrawal command to the control system. The effect is shown in Figure 8.

Experiments are performed on multiple sets of zebrafish embryo and microinjector tip micrographs. As illustrated in Table 2, the average accuracy of cell and microinjector tip detection at an IoU threshold of 0.5 is 99.5% and 97.3%, and at a threshold of 0.7, the accuracy is still 99.5% and 69.5%, respectively. The detection of cells, which are relatively large targets in the image, has a high success rate. In contrast, the detection of the microinjector tip, which belongs to a smaller target, has a lower success rate. However, the detection effect is sufficient to meet the experimental requirements. Compared to conventional image processing algorithms, a distinct advantage of this algorithm is that it eliminates the time-consuming process of progressively developing feature extraction algorithms

through multiple layers. Instead, the algorithm performs an end-to-end detection. This approach greatly enhances efficiency, necessitating only the preparation of the data set for model training at the outset. Following the completion of the model training, the microscopic images can be detected in real time. As illustrated in Table 3, the success rate for cell and microinjector tip detection on the test set is 100% and 94.1%, respectively. BI-YOLO demonstrates superior cellular detection performance compared to state-of-the-art methods in [30, 41, 42], achieving a higher success rate and faster inference speed on our microscopic manipulation dataset. Crucially, BI-YOLO integrates microinjector tip detection within its unified architecture, whereas existing approaches either require ancillary algorithmic components for microinjector tip localization or neglect microinjector tip localization entirely. This dual-functional design eliminates redundant feature extraction processes, resulting in fewer system-level latency spikes during continuous operation compared to multialgorithm solutions. The average detection speed for a single image is 118.8 FPS, which is essentially within 10 ms. Compared to other zebrafish detection algorithms, the BI-YOLO algorithm exhibits higher detection accuracy and faster detection speed. All experimental results for the proposed model presented in Tables 2 and 3 were obtained under strictly controlled hardware conditions: an Intel i7-13700H CPU paired with an NVIDIA GeForce RTX 3050 GPU. The evaluation was performed on a carefully

TABLE 4: Results of ablation experiments in the mAP at the different IoU threshold.

Model	mAP50		mAP60		mAP70		mAP80	
	Cell (%)	Microinjector tip (%)	Cell (%)	Microinjector tip (%)	Cell (%)	Microinjector tip (%)	Cell (%)	Microinjector tip (%)
YOLOv8n	99.5	96.8	99.5	86.0	99.5	60.6	98.8	29.1
YOLOv8n + DC	99.5	97.7	99.5	86.2	99.5	69.5	98.8	29.2
YOLOv8n + BiFPN	99.5	96.9	99.5	88.8	99.5	57.1	98.8	29.1
YOLOv8n + BiFPN + AFF	99.5	96.7	99.5	89.7	99.5	69.4	99.2	22.9
YOLOv8n + DC + BiFPN	99.5	95.8	99.5	85.7	99.5	67.4	98.9	25.5
BI-YOLO	99.5	97.8	99.5	90.2	99.5	69.5	99.2	32.3

Note: Bold values indicate the highest scores in each column, showing that the BI-YOLO performed best overall. Abbreviation: AFF, adaptive feature fusion.

TABLE 5: Results of ablation experiments in the parameters and GFLOPs.

Model	Parameters (M)	GFLOPs
YOLOv8n	3.00	8.1
YOLOv8n + DC	2.90	8.0
YOLOv8n + BiFPN	1.99	7.1
YOLOv8n + BiFPN + AFF	2.04	7.5
YOLOv8n + DC + BiFPN	1.88	7.0
BI-YOLO	1.96	7.2

curated test set used in Table 3 comprising 129 cell samples and 85 microinjector tip samples. The training in [31] was conducted using a GeForce RTX 2080 GPU, while Cordero-Maldonado et al. [42] employed a GeForce GTX 1070 GPU for training. Once the detection process is complete, the image coordinates of the cells and the microinjector tips are transformed into world coordinates and transmitted to the motion control module for microinjector movement and puncture. The duration required to complete individual injection procedures is consistently maintained below 0.5 s. Considering both the detection success rate and the operational efficiency of the automated microinjection system, the visual detection method proposed in this paper is undoubtedly the better choice.

4.2. Ablation Experiment. The detection results show that the BI-YOLO model has better detection of cells and microinjector tip, and also has better robustness in the face of the presence of obscuration in the environment. However, the specific enhancement of model detection performance by deformable convolution, BiFPN, and adaptive fusion is yet to be explored. Therefore, ablation studies are conducted on BI-YOLO to quantify the performance enhancements brought to the model by the three aforementioned improvements. The results of the ablation experiment for the BI-YOLO model are shown in Tables 4 and 5.

As illustrated in Table 4, with the IoU threshold increases, the mAP of the cells stabilizes at approximately 99%, while the mAP of the microinjector tip gradually decreases. Because the microinjector tip is a small object and the cells are large objects in the image. When the IoU threshold is set too high, the detection difficulty of the microinjector tip increases significantly. The inclusion of deformable convolutions in YOLOv8n

significantly improves the detection accuracy of the microinjector tip, especially with an 8.9% increase in mAP70. This demonstrates that deformable convolution enhance detection performance by adapting to objects of varying sizes. However, as shown in Table 5, the number of model parameters and computation is reduced slightly with the addition of deformable convolution. In consideration of the computational resources and subsequent model deployment, this paper introduces a lightweight BiFPN structure to further reduce the model parameters and computation cost. The traditional top-down FPN or bottom-up PAN structures are constrained by unidirectional information flow, which limits their accuracy. YOLOv8n employs the FPN and PAN strategy, resulting in enhanced accuracy. However, this approach necessitates additional parameters and GFLOPs. As illustrated in Tables 4 and 5, while BiFPN achieves slightly lower mAP70 compared to the FPN and PAN initially employed in YOLOv8n, it demonstrates a notable reduction in both parameters and FLOPs. With the incorporation of an adaptive feature fusion mechanism, the BiFPN has further enhanced its accuracy while maintaining a small number of parameters and computational cost. Finally, compared to YOLOv8n, BI-YOLO algorithm improves the mAP80 for cell detection by 0.4% and the mAP50 for microinjector tip detection increases by 1%, mAP60 by 4.2%, mAP70 by 8.9%, and mAP80 by 3.2%. Additionally, the model parameters decrease by 1.04 million, and GFLOPs are reduced by 0.9. The proposed model has been successfully deployed on an NVIDIA Jetson Nano embedded platform, demonstrating real-time detection capability at 35 FPS under continuous video streaming conditions.

4.3. Comparative Experiment. To thoroughly evaluate the ability of the BI-YOLO to detect cells and microinjector tips, this paper compares it with several popular target detection models in comparative tests. The results are shown in Table 6.

SSD [43] is a classic one-stage algorithm that initially outperformed YOLO, particularly in small target detection. However, over the years, YOLO has evolved to surpass SSD in both accuracy and efficiency. Although the RT_DETR [44] algorithm based on the attention [45] mechanism has been popular in the field of object detection in recent years, a significant gap remains between it and the YOLO series of one-stage algorithms in terms of hardware resource requirements and computing speed. While YOLOv9c exhibits superior outcomes in

TABLE 6: Performance comparison between BI-YOLO and popular target detection models.

Model	mAP50		mAP50-95		Parameters (M)	GFLOPs
	Cell (%)	Microinjector tip (%)	Cell (%)	Microinjector tip (%)		
SSD	100	94.9	89.4	28.9	23.70	60.9
YOLOv8n	99.5	96.8	94.6	50.3	3.00	8.1
YOLOv9c	99.5	97.8	95.4	52.1	25.33	102.3
YOLOv10n	99.5	95.8	95.0	48.8	2.70	8.2
RT_DETR	99.5	93.4	94.6	47.6	28.41	100.6
BI-YOLO	99.5	97.8	95.4	52.7	1.96	7.2

Note: Bold values indicate the highest scores in each column, showing that BI-YOLO performed best overall.

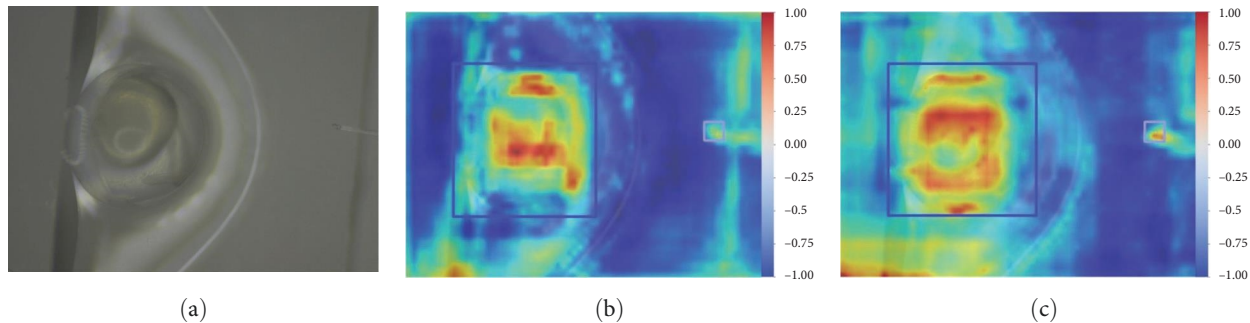


FIGURE 9: Class activation maps (CAM): (a) origin, (b) YOLOv8n, and (c) BI-YOLO.

comparison to YOLOv8n, its considerable number of parameters and the substantial amount of computation render it unsuitable for our lightweight requirements, particularly in view of the limited computational resources. Accordingly, in conjunction with the comprehensive comparison presented in Table 6, the BI-YOLO model developed in this paper demonstrates a high accuracy while maintaining a relatively low complexity, with only 1.96 million parameters and 7.2 GFLOPs of computation. The mAP50 values for cells and microinjector tips are 99.5% and 97.8%, respectively, while the mAP50-95 values are 95.4% and 52.7%, respectively. These results demonstrate that the algorithm meets the auto-injection, real-time detection, and localization requirements of an robotic injection system. Additionally, the lightweight nature of the model facilitates the subsequent deployment of models.

4.4. Class Activation Map. The class activation map is a visualization tool that allows the user to observe how the model responds to different regions within an image during a target detection task. This allows for an in-depth analysis of the underlying decision-making process of the model and the identification of potential areas for improvement [46, 47]. The heatmap's color scale encodes activation strengths: red (1) indicates strong positive correlation with the target class, blue (−1) signifies strong negative correlation (background class relevance), and white (0) denotes no significant correlation. As demonstrated by the comparative class activation maps in Figure 9, BI-YOLO achieves enhanced intracellular feature localization compared to YOLOv8n, with heatmap peaks densely clustered around cellular centroids, thereby extracting more discriminative morphological features. Furthermore, the architecture demonstrates robust responsiveness to

microinjector tips through its dual-target detection capability. Although marginally increased background activation is observed, this constitutes an acceptable trade-off given BI-YOLO's significant performance gains in both precision and efficiency, which collectively validate its superiority in automated cell microinjection systems.

4.5. Discussions. According to the aforementioned experimental verification, it can be demonstrated that the proposed deep learning-based real-time visual detection method for automated cell microinjection systems is effective and achieves high accuracy and efficiency in the detection of cells and microinjector tips. The primary reason for this is the introduction of deformable convolution and BiFPN with adaptive feature fusion, which enable the BI-YOLO algorithm to achieve efficient feature extraction with a minimum number of parameters and computation. Moreover, the simultaneous detection of both the cell and the microinjector tip, coupled with real-time positional feedback, serves to decrease the risk of cell damage caused by the microinjector tip. Subsequently, the algorithm can also be applied to scenarios in which there are significant size differences between targets within the FOV.

However, some limitations still exist in this work. First, the method is ineffective when the cells and microinjector tip are outside the microscope FOV, requiring manual adjustment of both the microinjector tip and the cells to bring them into view. Second, the microinjector tip, being a small target in the image, is susceptible to environmental influences. For instance, insufficient illumination can significantly impede the ability to detect the microinjector tip, potentially leading to missed detection. This can be mitigated by augmenting the capacity of detection algorithm to detect small targets. Finally, the

detection of two-dimensional images presents a challenge in obtaining information about the depth of the microinjector tip and the cell. This leads to difficulties in aligning the microinjector tip with the center of the cell, necessitating adjustments to the relative location of the microinjector tip and the cell from the beginning. Future studies may integrate vision-guided feedback control with existing force regulation to achieve submicron injection precision in cell micromanipulation systems [48].

5. Conclusion

This paper presents deep-learning based real-time visual detection method for robotic cell microinjection system with high accuracy and efficiency. This method introduces deformable convolution and BiFPN with adaptive feature fusion into the YOLOv8n algorithm, enabling the model to maintain high precision in targeting cells and microinjector tips while operating with fewer parameters and a lower computational load. Furthermore, the algorithm effectively identifies the microinjector tip in the presence of interference, demonstrating the robustness of the algorithm and its ability to provide consistent and reliable coordinate data and visual feedback for the subsequent motion control system. The integration of this method with movement implementation part alleviates the burden of repetitive and tedious injection procedures for experimenter. Moreover, it can provide efficient assistance for inexperienced users in zebrafish injections. In future work, the method collaborated with the motion control system will have the potential for application in robotic batch cell microinjection with high efficiency. Additionally, the injection force is determined by detecting the deformation degree of the cell membrane during injection. This injection force can then be fed back to the movement implementation part, creating a closed-loop mechanism that further enhances the performance of the microinjection system.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

Guozhi Liu and Shengzheng Kang contributed equally to this work.

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